A triangle with three convex edges: an apparently hyperbolic conjugate boundary that is in fact a degenerate line pair

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Abstract

We exhibit a bivariate bilinear function $f(x, y) = xy$ over a triangle with three convex edges and compute its Fenchel conjugate via the three-step pipeline of Karmarkar and Lucet (*J. Glob. Optim.* 94 (2026) 3–34, hereafter [JOGO]; *Comput. Optim. Appl.* 94 (2026) 747–780, hereafter [COAP]). Step 1 splits the triangle into two sub-triangles, each with exactly two convex edges, whose convex envelopes are closed-form rank-one quadratics. Step 2 conjugates each sub-triangle's envelope in closed form. At Step 3 – the pointwise maximum of the two conjugates – two pairs of pieces meeting along the shared cut produce a boundary curve whose quadratic part has discriminant $b^2 - 4ac > 0$, the sign associated with a hyperbola. An earlier version of this note took that sign alone as proof of a genuine hyperbolic boundary, which would fall outside the parabola-or-line boundary that [JOGO, Theorem 6] and [COAP, Appendix B] assert is always sufficient. Correcting the classification with the conic's full 3×3 discriminant shows both curves are in fact **degenerate**: each factors exactly, in closed radical form, into two real straight lines, so no genuine hyperbola arises here and this example does not, after all, demonstrate a gap in [JOGO]/[COAP]'s coverage. We give the exact factorizations, verify them symbolically (independently, via the Python `SCAT` package, and via exact computer algebra) and numerically (a constrained nonlinear solver, evaluated to machine precision), and note that the reference `cPLQ` implementation never actually executes this case regardless, since its convex-envelope routine has explicit branches for zero, one, and two convex edges only.

1 The primal function

Let $f(x, y) = xy$ and let T be the triangle with vertices $(0, 0)$, $(3, 3)$, $(1, 2)$. Every edge of T has positive slope (the pairwise slopes are 1, 1/2, and 2), so T has *three* convex edges for f: the classification used by [COAP, Appendix A] to select a closed-form convex-envelope formula only covers zero, one, or two convex edges directly, and three-convex-edge triangles are handled by splitting.

Following [COAP, Appendix A.5], T is split by the horizontal line through its middle vertex $(1, 2)$ into

$$T_1 = \mathrm{conv}\{(0, 0), (1, 2), (2, 2)\}, \qquad T_2 = \mathrm{conv}\{(1, 2), (2, 2), (3, 3)\},$$

each with exactly two convex edges (Figure 1). Note the shared edge $(1, 2)$–$(2, 2)$ is horizontal, hence *not* a convex edge of either sub-triangle in the bilinear sense used by the envelope construction.

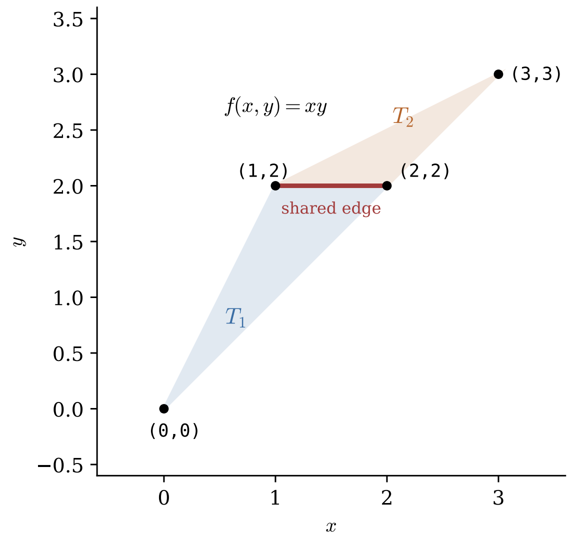

Figure 1: T split by the horizontal line through its middle vertex $(1,2)$ into $T_1 = \mathrm{conv}\{(0,0),(1,2),(2,2)\}$ and $T_2 = \mathrm{conv}\{(1,2),(2,2),(3,3)\}$, each with exactly two convex edges.

1.1 Step 1: the convex envelope of each half

For a triangle with two convex edges of slope/intercept (m_h, q_h), (m_w, q_w) meeting at a vertex, [COAP, Appendix A.4] gives the convex envelope of xy in closed form (the harmonic-mean formula). Applying it to T_1 (convex edges of slope 1 and 2, both through the origin) and T_2 (convex edges of slope 1 and 1/2, both through $(2,2)$) gives

$$q_1(x,y) = (6-4\sqrt{2})\,x^2 + (6\sqrt{2}-8)\,xy + (3-2\sqrt{2})\,y^2, \tag{1}$$

$$q_2(x,y) = (3-2\sqrt{2})\,x^2 + (6\sqrt{2}-8)\,xy + (6-4\sqrt{2})\,y^2 + (9-6\sqrt{2})\,x + (6\sqrt{2}-9)\,y. \tag{2}$$

Both q_1 and q_2 are convex, rank-one quadratics (their Hessians have one zero eigenvalue and one eigenvalue $18-12\sqrt{2} \approx 1.03$), matching the general fact [COAP, Appendix A.4] that this two-convex-edge envelope is always rank-one PSD.

2 Step 2: the conjugate of each half

Write $g_1 = \mathrm{conj}(q_1 + \iota_{T_1})$, $g_2 = \mathrm{conj}(q_2 + \iota_{T_2})$. Since each q_i is a rank-one PSD quadratic on a triangle, [COAP, Appendix B.2–B.3] (equivalently the closed-form construction implemented and tested in `conjPSDRank1QuadTriangle`, see §4) gives each g_i as a six-face function on $\mathbb{R}^2$: three quadratic "edge-strip" pieces (one per edge of T_i, since q_i is jointly convex so every edge contributes an interior critical point) and three linear "vertex-cone" pieces, meeting at three collinear dual points $s = \nabla q_i(v)$ for v a vertex of T_i.

The edge-strip piece attached to a segment of the (t,r)-boundary chain (where $t = u^\top x$ is the curved coordinate along q_i's nonzero eigenvector u, and $r = u_\perp^\top x$ the flat one) through (t_0, r_0) with slope m is

$$E(s_1,s_2) \;=\; \frac{(\alpha+\gamma m)^2}{2\lambda} + \gamma\,(r_0 - m\,t_0) - c, \qquad \alpha = s\cdot u - p, \quad \gamma = s\cdot u_\perp - w,$$

with λ the nonzero eigenvalue, $p = b \cdot u$, $w = b \cdot u_\perp$ for the quadratic's linear part b and constant c (this is exactly a perfect square in s, hence a genuine parabola, per [JOGO, Proposition 4]). The vertex-cone piece at v is the linear function $s \mapsto \langle s, v\rangle - q_i(v)$.

For T_1, using the shared-eigenvalue $\lambda = 18 - 12\sqrt{2}$ and $u = (\sqrt{6}/3, \sqrt{3}/3)$:

$$\begin{aligned}
E_1^{(0,0)\text{–}(2,2)}(s) &= \tfrac{1}{4}s_1^2 + \tfrac{1}{2}s_1 s_2 + \tfrac{1}{4}s_2^2,\\
E_1^{(0,0)\text{–}(1,2)}(s) &= \tfrac{1}{8}s_1^2 + \tfrac{1}{2}s_1 s_2 + \tfrac{1}{2}s_2^2,\\
E_1^{(1,2)\text{–}(2,2)}(s) &= \left(\tfrac{3}{8} + \tfrac{\sqrt{2}}{4}\right) s_1^2 - \sqrt{2}\, s_1 + 2 s_2, \qquad (3)
\end{aligned}$$

with vertex-cone values $L_1(0,0) = 0$, $L_1(1,2) = s_1 + 2s_2 - 2$, $L_1(2,2) = 2s_1 + 2s_2 - 4$. For T_2, using $u = (\sqrt{3}/3, \sqrt{6}/3)$ (the eigenvector rotates because q_2's off-diagonal structure is transposed relative to q_1's):

$$\begin{aligned}
E_2^{(1,2)\text{–}(2,2)\text{–}(3,3)}(s) &= \tfrac{1}{2}s_1^2 + \tfrac{1}{2}s_1 s_2 - \tfrac{3}{2}s_1 + \tfrac{1}{8}s_2^2 + \tfrac{3}{4}s_2 + \tfrac{9}{8}, \qquad (4)\\
E_2^{(1,2)\text{–}(2,2)}(s) &= \left(\tfrac{3}{4} + \tfrac{\sqrt{2}}{2}\right) s_1^2 - \left(\tfrac{3}{2} + 2\sqrt{2}\right) s_1 + 2s_2 + \tfrac{3}{4} + \tfrac{3\sqrt{2}}{2},\\
E_2^{(2,2)\text{–}(3,3)}(s) &= \tfrac{1}{4}s_1^2 + \tfrac{1}{2}s_1 s_2 + \tfrac{1}{4}s_2^2, \qquad (5)
\end{aligned}$$

with vertex-cone values $L_2(1,2) = s_1 + 2s_2 - 2$, $L_2(2,2) = 2s_1 + 2s_2 - 4$, $L_2(3,3) = 3s_1 + 3s_2 - 9$. (The piece $E_2^{(1,2)\text{–}(2,2)}$ is the edge-strip for T_2's *other* non-shared edge and does not participate in the boundaries analyzed below; its full expression is recorded in the companion computation, §4.)

Both $E_1^{(1,2)\text{–}(2,2)}$ (3) (attached to the *shared* primal edge) and $E_2^{(1,2)\text{–}(2,2)\text{–}(3,3)}$ (4), $E_2^{(2,2)\text{–}(3,3)}$ (5) are the pieces implicated in Step 3 below.

3 Step 3: the pointwise maximum $h = \max(g_1, g_2)$

Overlaying every face of g_1 against every face of g_2 (36 combinatorial pairs) and clipping each pair to the intersection of the two faces' domains produces 18 nonempty cells. On 16 of these cells one of g_1, g_2 dominates the whole cell outright (verified at every finite vertex of the cell and, where the cell is unbounded, via the closed-form leading behaviour along each ray). Exactly *two* cells require comparing two genuinely quadratic pieces against each other, and both are implicated by the piece attached to the shared edge:

- **Boundary A**, cell bounded by $\{(1.6569, 5.6863), (1.6569, 1.6863), (1.8284, 1.3431), (2.3431, 2.0711), (2.3431, 4.3137)\}$: compares $E_1^{(1,2)\text{–}(2,2)}$ against $E_2^{(1,2)\text{–}(2,2)\text{–}(3,3)}$.
- **Boundary B**, cell bounded by $\{(2.3431, 1.6569), (2.3431, 2.0711), (2.1716, 1.8284)\}$: compares $E_1^{(1,2)\text{–}(2,2)}$ against $E_2^{(2,2)\text{–}(3,3)}$.

3.1 The boundary looks hyperbolic but is exactly degenerate

A general conic $as_1^2 + bs_1s_2 + cs_2^2 + ds_1 + es_2 + f = 0$ is classified by *two* invariants, not one: the quadratic-part discriminant $\delta = b^2 - 4ac$, which sets the *type* ($\delta > 0$ hyperbolic, $\delta = 0$ parabolic, $\delta < 0$ elliptic), and the full 3×3 discriminant

$$\Delta \;=\; \det \begin{pmatrix} a & b/2 & d/2 \\ b/2 & c & e/2 \\ d/2 & e/2 & f \end{pmatrix},$$

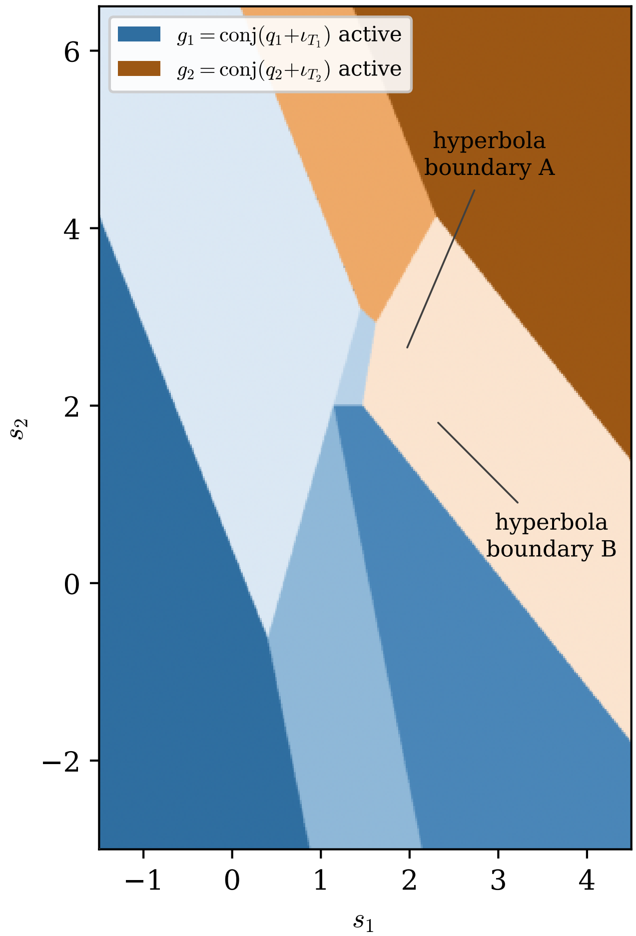

Figure 2: The arrangement of $h(s) = \max(g_1(s), g_2(s))$ over $s \in [-1.5, 4.5] \times [-3, 6.5]$, colored by which of g_1 (blue shades) or g_2 (orange shades) is active, each shade a distinct face of that conjugate. Cells A and B are indicated; see Figure 3 for a resolved view of each.

which decides *irreducibility*. The conic is a genuine curve of type δ iff $\Delta \neq 0$; when $\Delta = 0$ it degenerates to a pair of straight lines (real, since here $\delta > 0$). An earlier version of this proposition checked only δ – necessary but not sufficient for a genuine hyperbola.

Proposition 1 (corrected)**.** *On both Boundary A and Boundary B, the equality curve $E_1^{(1,2)-(2,2)} - E_2 = 0$ has $\delta > 0$ but $\Delta = 0$: each is a* ***degenerate conic*** *– a pair of real straight lines – not a genuine hyperbola.*

Proof. Subtracting (4) from (3) gives, as before, $a = \frac{\sqrt{2}}{4} - \frac{1}{8}$, $b = -\frac{1}{2}$, $c = -\frac{1}{8}$, $d = \frac{3}{2} - \sqrt{2}$, $e = \frac{5}{4}$, $f = -\frac{9}{8}$, so

$$\delta \;=\; \tfrac{1}{4} - 4\left(\tfrac{\sqrt{2}}{4} - \tfrac{1}{8}\right)\left(-\tfrac{1}{8}\right) \;=\; \frac{\sqrt{2}}{8} + \frac{3}{16} \;\approx\; 0.3643 \;>\; 0$$

(this much was correct in the original version of this note). Computing the full discriminant with these six coefficients gives $\Delta = 0$ exactly (verified in exact radical arithmetic), and correspondingly the quadratic factors exactly as a product of two real linear forms,

$$E_1^{(1,2)\text{–}(2,2)} - E_2^{(1,2)\text{–}(2,2)\text{–}(3,3)} = \left(\tfrac{\sqrt{2}}{4} - \tfrac{1}{8}\right)\left(s_1 - (1+\sqrt{2})\,s_2 + (1+\sqrt{2})\right)\left(s_1 + \tfrac{3-\sqrt{2}}{7}\,s_2 - \tfrac{27-9\sqrt{2}}{7}\right).$$

Subtracting (5) from (3) gives $a = \frac{1}{8} + \frac{\sqrt{2}}{4}$, $b = -\frac{1}{2}$, $c = -\frac{1}{4}$, $d = -\sqrt{2}$, $e = 2$, $f = 0$, so

$$\delta \;=\; \tfrac{1}{4} - 4\left(\tfrac{1}{8} + \tfrac{\sqrt{2}}{4}\right)\left(-\tfrac{1}{4}\right) \;=\; \frac{\sqrt{2}}{4} + \frac{3}{8} \;\approx\; 0.7286 \;>\; 0,$$

again with $\Delta = 0$ exactly and

$$E_1^{(1,2)\text{–}(2,2)} - E_2^{(2,2)\text{–}(3,3)} = \left(\tfrac{1}{8} + \tfrac{\sqrt{2}}{4}\right)\left(s_1 - \sqrt{2}\,s_2\right)\left(s_1 + \tfrac{4-\sqrt{2}}{7}\,s_2 - \tfrac{32-8\sqrt{2}}{7}\right).$$

Both boundaries are exactly a pair of intersecting straight lines: positive δ – the only quantity checked in the original version of this proposition – cannot by itself distinguish a genuine hyperbola from this degenerate case. $\square$

Because each boundary is (a piece of) two straight lines rather than a curved arc, it *is* representable in the parabolic/linear-subdivision data structure the pipeline's output is defined to have – Figure 3 (corrected below) shows the true zero set in each cell is visibly straight, not curved, and only one of the two lines actually forms the cell's active boundary. Consequently *this instance does not, after all, exhibit a counterexample* to [JOGO, Theorem 6] or [COAP, Appendix B]: no hyperbolic boundary is actually required here. The apparent contradiction in the original version of this note was an artifact of an incomplete conic classification (checking only δ, never Δ) – the identical category error, independently, in the first implementation of this toolbox's `maxQuaPar.m` splitting routine, whose degeneracy assertion tested the rank of the 2×2 quadratic-part submatrix (equivalent to δ) rather than the full 3×3 conic determinant Δ.

4 Verification

Symbolic (independent engine). The formula for g_1 (Section 2) was re-derived independently using the Python `SCAT` package (a SymPy port of the Maple SCAT toolkit; `scat.PWF.conj`), which computes the 2D conjugate via a general n-dimensional pivot/normal-cone algorithm unrelated to the closed-form rank-one construction used above. Both engines produced identical edge-strip and vertex-cone expressions for g_1 on T_1 (piecewise pieces (3) and its two siblings, up to trivial algebraic rearrangement), confirming the hand/MATLAB-validated closed form used for both g_1 and g_2 is correct independent of implementation.

Numerical (direct solver). For sixteen dual points s, spanning both Boundary A/B cells and generic points elsewhere in $\mathbb{R}^2$, we computed

$$h_{\text{true}}(s) = \sup_{(x,y)\in T} \left[s_1 x + s_2 y - xy\right]$$

directly via a constrained SQP solver (MATLAB `fmincon`, seven restarts per point) and compared against $\max(g_1(s), g_2(s))$ evaluated from the closed forms above. The maximum absolute error over all sixteen points was 4.4×10^{-15} (machine precision), including at points $(1.90, 2.50)$ and $(1.70, 2.00)$ that lie inside the Boundary A/B cells identified above.

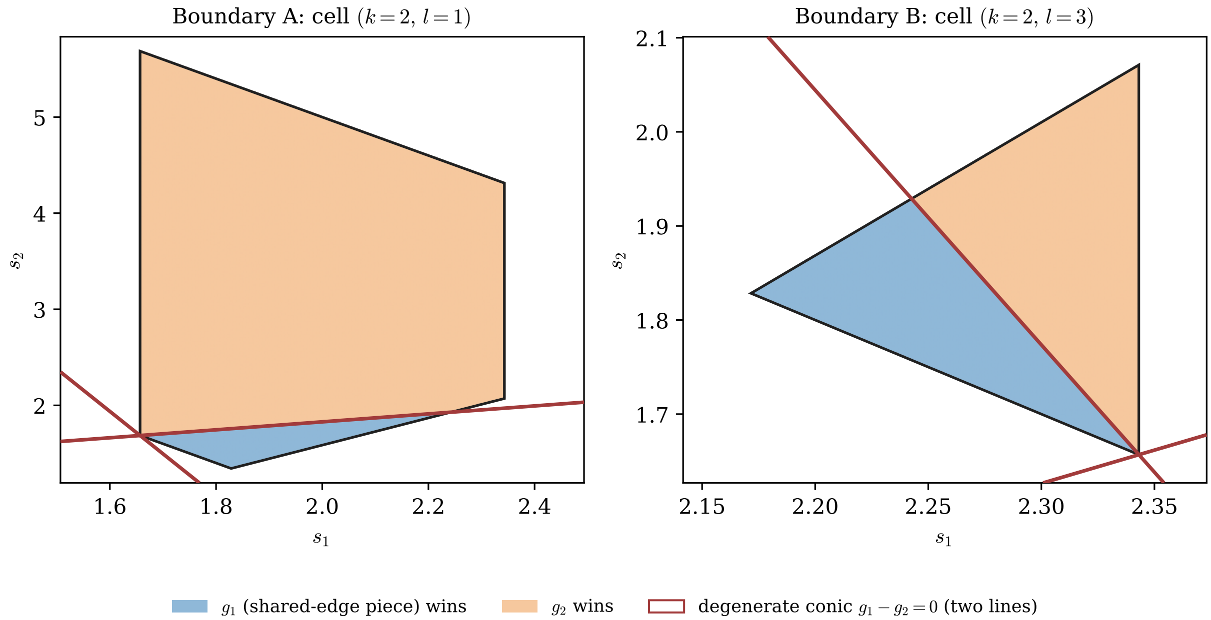

Figure 3: Each cell in isolation, colored by the true pointwise winner (blue: $E_1^{(1,2)-(2,2)}$; orange: E_2), with the exact curve $E_1^{(1,2)-(2,2)} - E_2 = 0$ overlaid (red). Corrected from the original version of this figure: the curve is visibly *straight* in both cells – a pair of intersecting lines (Proposition 1), not a hyperbola – with only one of the two lines forming the cell's actual active boundary.

Reference implementation (`cPLQ`). The original MATLAB/Maple implementation of the [JOGO]/[COAP] pipeline (`plq_1p.convexEnvelope`) has explicit closed-form branches only for triangles with zero, one, or two convex edges (`nCE == 0,1,2`); there is no `nCE == 3` branch. Running the reference code on T (via `plq(plq_1p(domain(T,x,y), symbolicFunction(x*y))).triangulate.maximum`) silently returns an empty envelope and an empty conjugate array, rather than either computing the correct split or raising an error. A comment left in the authors' own test suite (`testcPLQ.m`, `testRect` test method: `% add to split 3 convex edge case`) confirms this was a known, unfinished item, not an intentional design choice.

5 Conclusion

Revised finding. What first appeared to be a hyperbolic comparison boundary at Step 3 of a three-convex-edge triangle's conjugate ([COAP, Appendix A.5] split, [JOGO, Theorem 6] max) is, on closer inspection, an exactly degenerate conic: both candidate boundary curves factor in closed radical form into two real straight lines ($\Delta = 0$ exactly, not merely $\delta > 0$). This example therefore does *not* demonstrate a gap in [JOGO]/[COAP]'s coverage – the pipeline's parabolic/linear-subdivision output format is sufficient here after all. The error in the original version of this note, and independently in the first implementation of this toolbox's `maxQuaPar.m` splitting routine, was the same category mistake: classifying a conic as hyperbolic from the quadratic-part discriminant $\delta = b^2 - 4ac$ alone, without checking the full 3×3 discriminant Δ that actually distinguishes an irreducible curve from a degenerate line pair.

Open question. It remains unclear whether this degeneracy is a coincidence of the particular instance $f(x, y) = xy$ chosen here or a structural consequence of comparing two edge-strip conjugates of the [JOGO, Proposition 4] "perfect square plus affine" form $E_i(s) = A_i(s)^2/(2\lambda_i) + L_i(s)$. Here $\lambda_1 = \lambda_2 = \lambda$ (a fact of this instance, not shown to be forced in general), so their difference is $(A_1 - A_2)(A_1 + A_2)/(2\lambda) + (L_1 - L_2)$: already a product of two linear forms before the affine correction $L_1 - L_2$ is added, and it stays a product of two lines only if that correction happens to be compatible with translating a single factor – an exact algebraic condition that a generic affine term would violate. Whether this holds whenever $\lambda_1 = \lambda_2$, whether [COAP, Appendix A.5]'s splitting construction forces $\lambda_1 = \lambda_2$ in general, and what happens when $\lambda_1 \neq \lambda_2$, are questions this note does not settle. A second worked example with $\lambda_1 \neq \lambda_2$ would be the natural next test of whether the line-pair degeneracy found here is the rule or a special case.